Restaurant Analytics

Restaurant Sales Analysis

Exploring Restaurant Sales & Performance

Presented by
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Project Overview

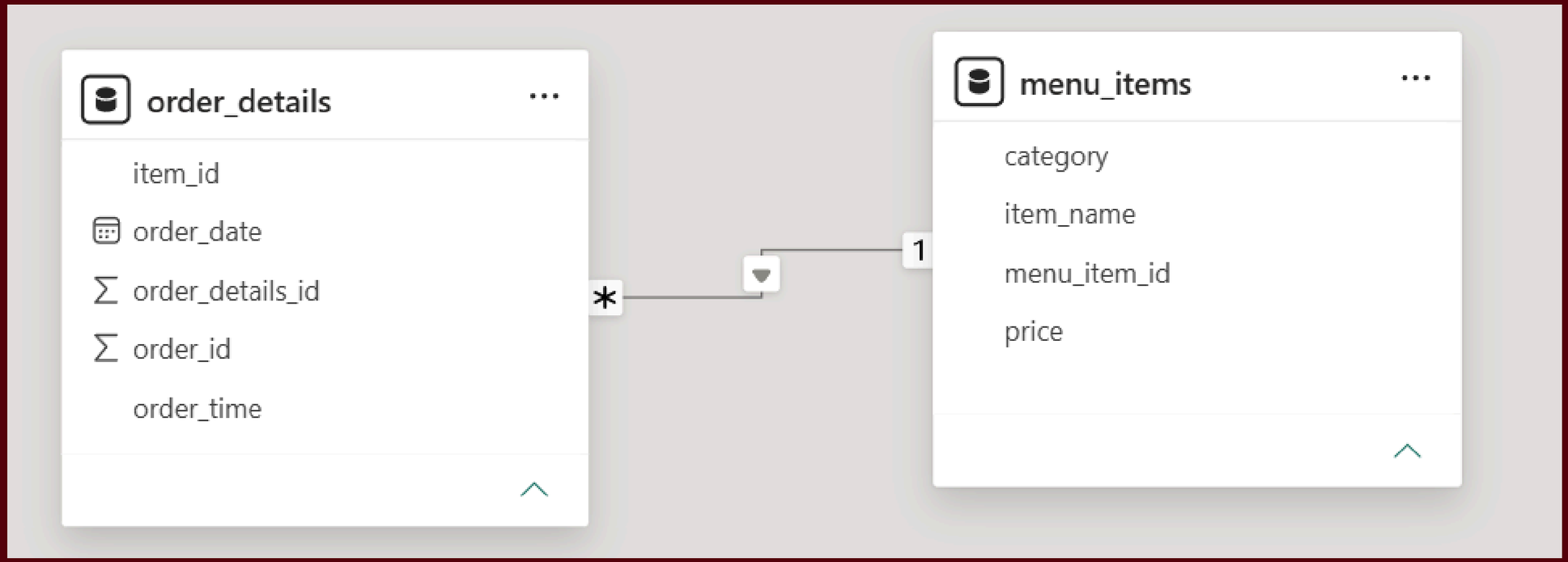
Hello everyone, I'm Akash Saraswat. This is my SQL project, for which I downloaded the Restaurant Orders dataset from Maven Analytics. After downloading the CSV files, I imported them into MySQL Workbench for analysis.

As part of my learning, I was given a set of business questions that required me to analyze the data and derive meaningful insights. The project consists of two tables: order_details and menu_items, with menu_item_id serving as the primary key in the menu_items table and item_id acting as the foreign key in order_details.

Through this project, I analyzed restaurant orders, revenue, menu-item performance, category-wise sales, daily and monthly trends, and ordering patterns across different hours. The project includes 11 questions covering basic, intermediate, and advanced SQL concepts.

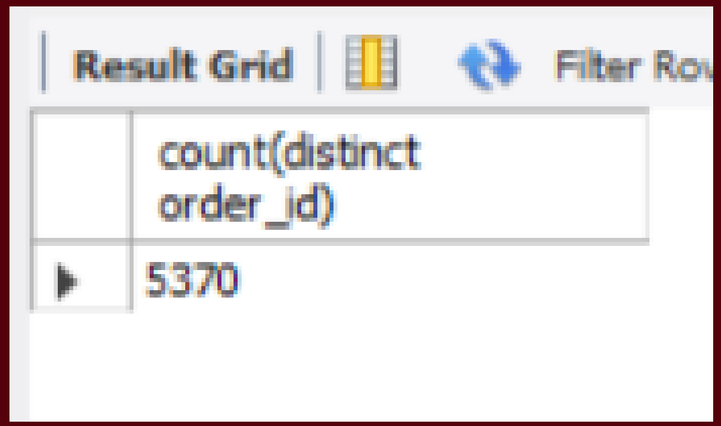
The objective was not just to write SQL queries but to use SQL to answer practical business questions and understand how data can support better business decisions.

Database Schema



1- How many unique orders did the restaurant receive during the entire period?

```sql
Select count(distinct order_id)
from order_details;
```



2- Give me a list of all menu items along with their category and price. Also, tell me how many items we have in each category?

```sql
Select
    item_name,
    category,
    price
From menu_items;

SELECT
    category,
    COUNT(menu_item_id) As items_each_category
FROM menu_items
GROUP BY category;
```

Result Grid | Filter Rows:

category	items_each_category
American	6
Asian	8
Mexican	9
Italian	9

3- Which are our 5 most expensive menu items? Show the item name, category, and price ?

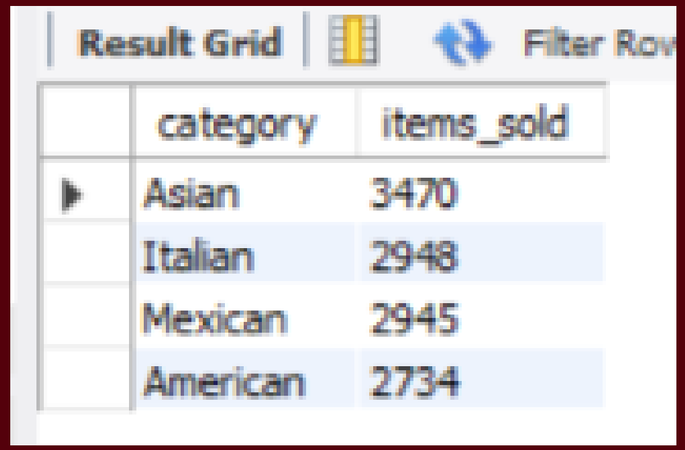
```sql
SELECT
    item_name, category, price
FROM
    menu_items
ORDER BY price DESC
LIMIT 5;
```

Result Grid | Filter Rows:

item_name	category	price
Shrimp Scampi	Italian	19.95
Korean Beef Bowl	Asian	17.95
Pork Ramen	Asian	17.95
Spaghetti & Meatballs	Italian	17.95
Meat Lasagna	Italian	17.95

4- How many items have we sold from each food category? I want to know which categories are most popular ?

```sql
SELECT
    menu_items.category,
    Count(order_details.item_id) As items_sold
From order_details
JOIN menu_items
    on order_details.item_id = menu_items.menu_item_id
group by menu_items.category
Order by items_sold DESC;
```



category	items_sold
Asian	3470
Italian	2948
Mexican	2945
American	2734

5- Which 10 menu items have been ordered the most? Show the item name and total quantity sold ?

```sql
Select
    menu_items.item_name,
    COUNT(order_details.item_id) As total_quantity_sold
FROM order_details
Join menu_items
    ON order_details.item_id = menu_items.menu_item_id
Group by menu_items.item_name
Order by total_quantity_sold desc
Limit 10;
```

Result Grid | Filter Rows:

item_name	total_quantity_sold
Hamburger	622
Edamame	620
Korean Beef Bowl	588
Cheeseburger	583
French Fries	571
Tofu Pad Thai	562
Steak Torta	489
Spaghetti & Meatballs	470
Mac & Cheese	463
Chips & Salsa	461

6- How much revenue has each food category generated? Rank the categories from highest to lowest revenue ?

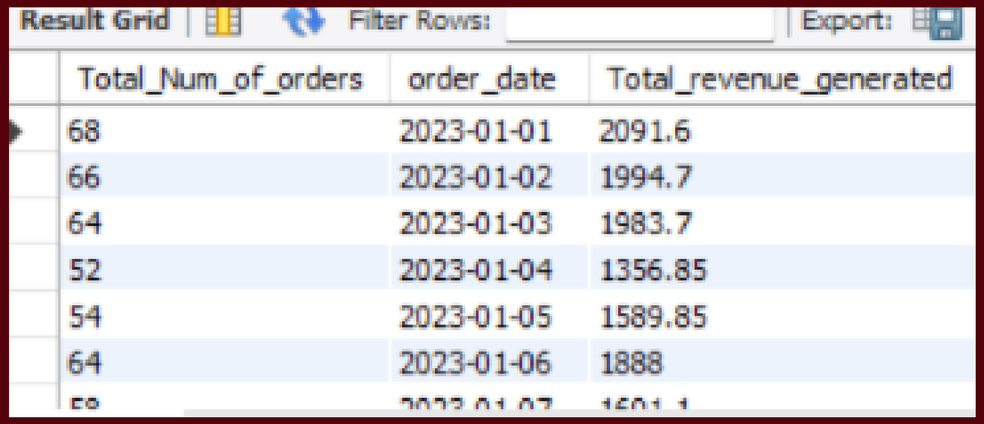
```sql
SELECT
    menu_items.category,
   Round( SUM(menu_items.price),2) AS revenue_generated
FROM menu_items
JOIN order_details
    ON menu_items.menu_item_id = order_details.item_id
GROUP BY menu_items.category
ORDER BY revenue_generated DESC;
```

Result Grid | Filter Rows:

category	revenue_generated
Italian	49462.7
Asian	46720.65
Mexican	34796.8
American	28237.75

7- Show me the total number of orders and total revenue generated on each day. I want to understand how our sales are performing day by day ?

```sql
Select
    COUNT(DISTINCT order_details.order_id) AS Total_Num_of_orders,
     order_details.order_date,
    ROUND (SUM(menu_items.price),2) AS Total_revenue_generated
From
    order_details
        JOIN
    menu_items ON order_details.item_id = menu_items.menu_item_id
GROUP BY order_details.order_date
Order by order_details.order_date;
```



Result Grid | Filter Rows: | Export:

Total_Num_of_orders	order_date	Total_revenue_generated
68	2023-01-01	2091.6
66	2023-01-02	1994.7
64	2023-01-03	1983.7
52	2023-01-04	1356.85
54	2023-01-05	1589.85
64	2023-01-06	1888
58	2023-01-07	1601.1

8- At what times of the day are customers ordering the most? Break the orders down by hour and identify our busiest hours ?

```sql
Select
    HOUR(order_time) As order_hour,
    COUNT(DISTINCT order_id) As total_orders
FROM order_details
GROUP BY HOUR(order_time)
ORDER BY order_hour desc;
```

Result Grid | Filter Rows:

order_hour	total_orders
15	357
14	426
13	595
12	647
11	288
10	2

9- Which menu items have generated more than $2,000 in total revenue? Show the item name, category, quantity sold, and revenue ?

```sql
SELECT
    menu_items.item_name,
    menu_items.category,
    COUNT(order_details.item_id) AS quantity_sold,
    Round(SUM(menu_items.price),2) AS total_revenue
FROM menu_items
Join order_details
    ON menu_items.menu_item_id = order_details.item_id
GROUP BY
    menu_items.item_name,
    menu_items.category
Having SUM(menu_items.price) > 2000
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export:

item_name	category	quantity_sold	total_revenue
Korean Beef Bowl	Asian	588	10554.6
Spaghetti & Meatballs	Italian	470	8436.5
Tofu Pad Thai	Asian	562	8149
Cheeseburger	American	583	8132.85
Hamburger	American	622	8054.9

10- Identify the top 3 most-ordered menu items within each food category, based on the number of times each item was ordered ?

```sql
WITH item_sales AS
(Select menu_items.category,
        menu_items.item_name,
        COUNT(order_details.item_id) AS quantity_sold
    From menu_items
    Join order_details
        ON menu_items.menu_item_id = order_details.item_id
    Group by
        menu_items.category,
        menu_items.item_name),
ranked_items AS
(SELECT category, item_name, quantity_sold,
        RANK() OVER (
            Partition by category
            ORDER BY quantity_sold DESC
        ) AS item_rank
    FROM item_sales )
SELECT category, item_name, quantity_sold, item_rank
From ranked_items
Where item_rank <= 3
Order by category, item_rank;
```

Result Grid | Filter Rows: | Export: | Wr

category	item_name	quantity_sold	item_rank
American	Hamburger	622	1
American	Cheeseburger	583	2
American	French Fries	571	3
Asian	Edamame	620	1
Asian	Korean Beef Bowl	588	2
Asian	Tofu Pad Thai	562	3
Italian	Spaghetti & Meatballs	470	1
Italian	Eggplant Parmesan	420	2
Italian	Spaghetti	367	3
Mexican	Steak Torta	489	1
Mexican	Chips & Salsa	461	2
Mexican	Chicken Burrito	455	3

11- Calculate the total revenue for every month and rank the months from highest to lowest based on revenue. Which month performed the best ?

```
WITH monthly_revenue As
(Select Month(order_date) AS order_month,
        SUM(menu_items.price) AS total_revenue
    From order_details
    JOIN menu_items
        ON order_details.item_id = menu_items.menu_item_id
    Group by Month(order_date))
SELECT order_month,
    ROUND(total_revenue, 2) AS total_revenue,
    RANK()  Over(ORDER BY total_revenue desc) AS revenue_rank
From monthly_revenue
ORDER BY revenue_rank;
```

Result Grid | Filter Rows:

order_month	total_revenue	revenue_rank
3	54610.6	1
1	53816.95	2
2	50790.35	3

Project Conclusion

Key Insights

1- Italian Category Leads Revenue
Italian cuisine generated the highest category revenue, followed by Asian, Mexican, and American categories.

2- High-Performing Menu Items Drive Revenue
Several menu items crossed $2,000 in total revenue, with Korean Beef Bowl generating the highest revenue-generating item among them.

3- Lunch Hours See the Highest Demand
Customer ordering activity peaks around 12 PM, making lunchtime the busiest period for the restaurant.

4- Menu Performance Varies Across Categories
Top-performing items vary across categories, showing that each category has its own best-selling menu items.

5- Revenue Performance Over Time
Daily and monthly analysis shows how the restaurant's revenue changes over time and highlights the periods with higher and lower sales.